

ML automation with **R²OBERT** AUTOMATED ML PROTOCOLS

David Dalmau

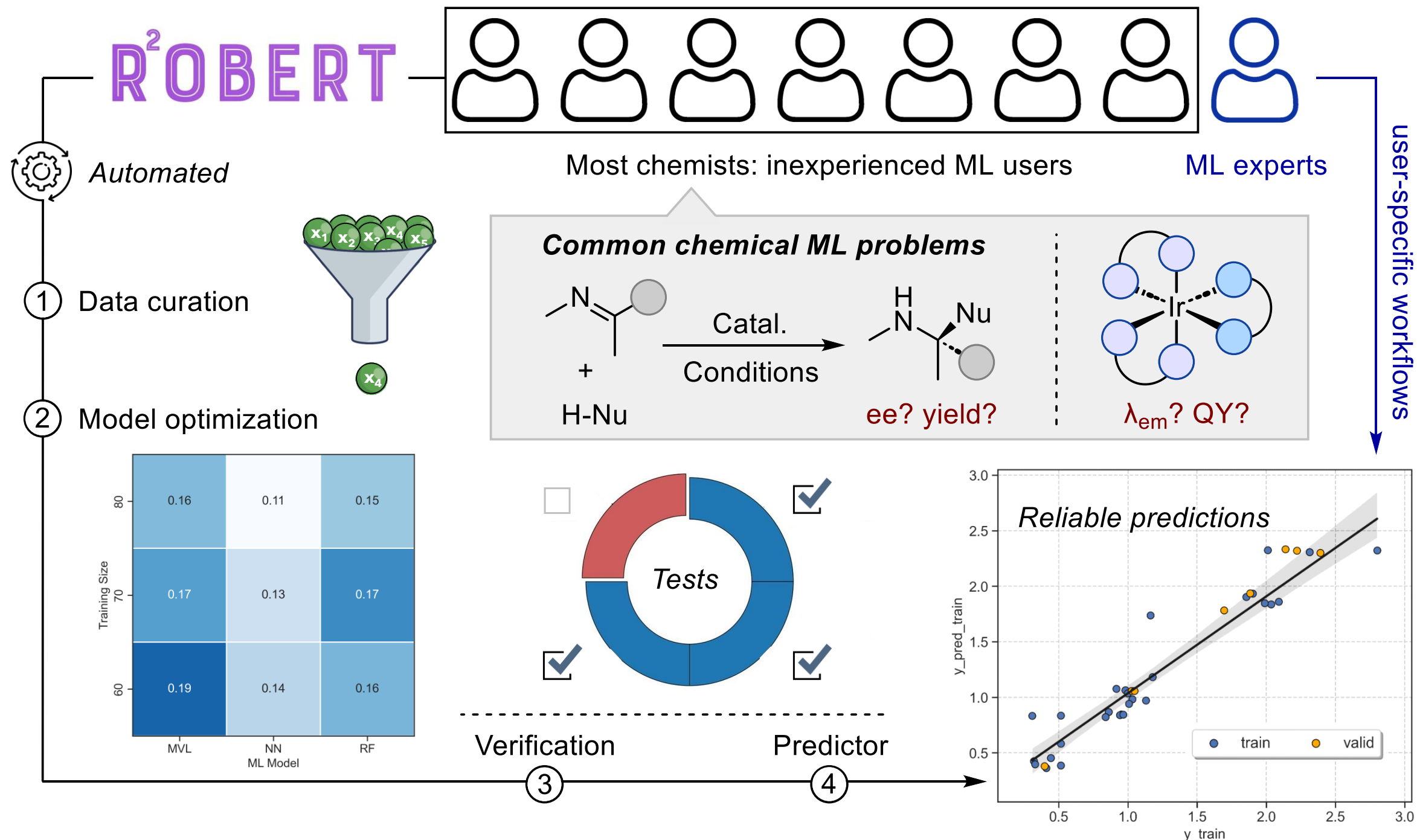
Workflow: 04/06/2026



Introduction



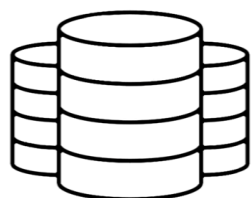
R²OBERT
AUTOMATED ML PROTOCOLS



ROBERT: automation of ML protocols



R²OBERT
AUTOMATED ML PROTOCOLS



■ CSV database

Input: CSV file with database or SMILES

User-defined descriptors

— X_i — — y —

code_name	charge	LUMO	...	solub.
mol_5	0	-6,581	...	-13,3
mol_13	0	-5,375	...	-2,68
mol_16	0	-4,349	...	-1,41
mol_34	0	-6,557	...	-1,64

or

Automated: SMILES to descriptors

— y —

SMILES	solub.
c1ccsc1	-13,3
CCCC=C	-2,68
CC(C)Cl	-1,41
ClC(=C)Cl	-1,64

— X_i —

- Charge
- χ_n
- FOD
- V_{bur}
- ...

200+ descriptors
(RDKit, xTB & DBSTEP)

Executing ROBERT: only one command line needed



```
python -m robert --csv_name FILENAME.csv --y y_NAME (--ARG1 --ARG2 ...)
```



① Data curation ② Model optimization ③ Verification ④ Predictor

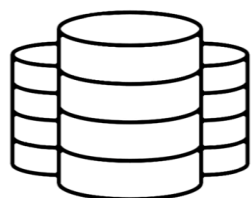
- ✓ Data curation
- ✓ Statistical tests
- ✓ Outlier analysis
- ✓ ML model scan
- ✓ Predictor model
- ✓ AQME tandem

Full workflow in one command line

ROBERT: automation of ML protocols



R²OBERT
AUTOMATED ML PROTOCOLS



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Automated: SMILES to descriptors

	y		X_i
SMILES	solub.		
<chem>c1ccsc1</chem>	-13,3	■ Charge	■ FOD
<chem>CCCC=C</chem>	-2,68	■ Chi _n	■ V _{bur}
<chem>CC(C)Cl</chem>	-1,41	...	
<chem>ClC(=C)Cl</chem>	-1,64	200+ descriptors (RDKit, xTB & DBSTEP)	

or

Executing ROBERT: only one command line needed



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```

- ✓ Data curation
- ✓ ML model scan
- ✓ Statistical tests
- ✓ Predictor model
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- ✓ AQME tandem



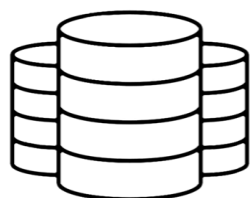
- ① Data curation
- ② Model optimization
- ③ Verification
- ④ Predictor

Full workflow in one command line

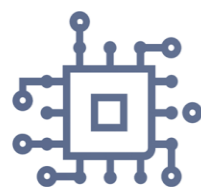
ROBERT: automation of ML protocols



R²OBERT
AUTOMATED ML PROTOCOLS



■ CSV database



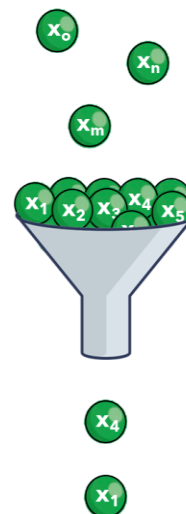
R²OBERT
AUTOMATED ML PROTOCOLS

- ✓ Data curation
- ✓ Statistical tests
- ✓ Outlier analysis
- ✓ ML model scan
- ✓ Predictor model
- ✓ AQME tandem

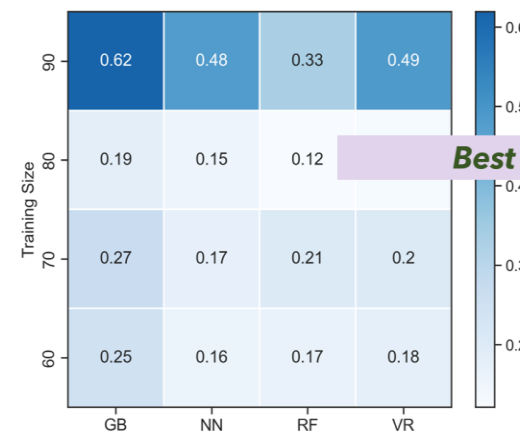
CURATE

Data curation process

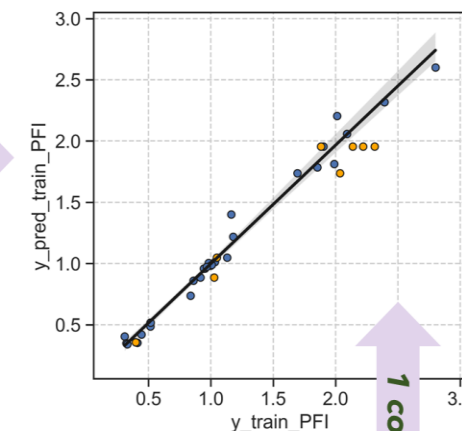
- Discard noise
- Discard correlated descriptors
- Categorical data conversion
- Duplicate filter



GENERATE

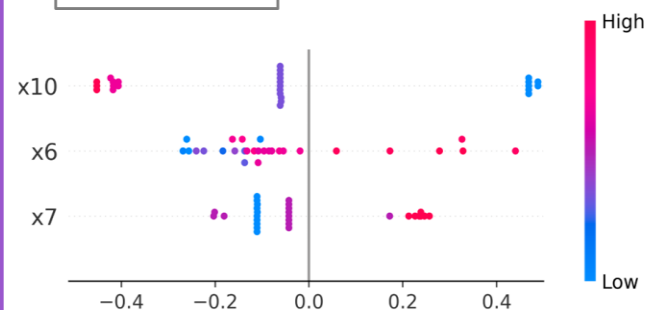


■ ML model and partition screening



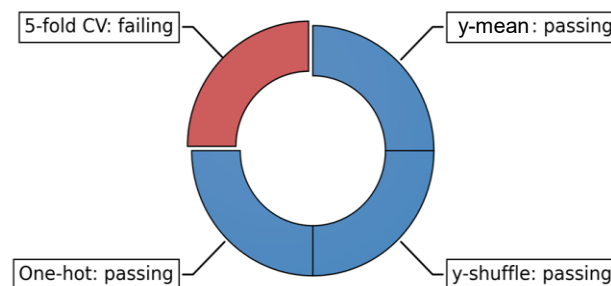
■ PFI filtering

PREDICT



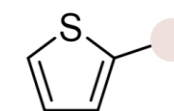
■ External test set ■ SHAP Analysis

VERIFY

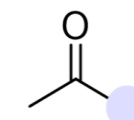


■ Statistical test analysis

AQME



C1=CC=CS1
C1C1=CC=CS1



CC(C)=O
CC(C(C)NC)=O

■ SMILES to results

Full workflow in one command line

CURATE

1. Filters off **correlated descriptors**

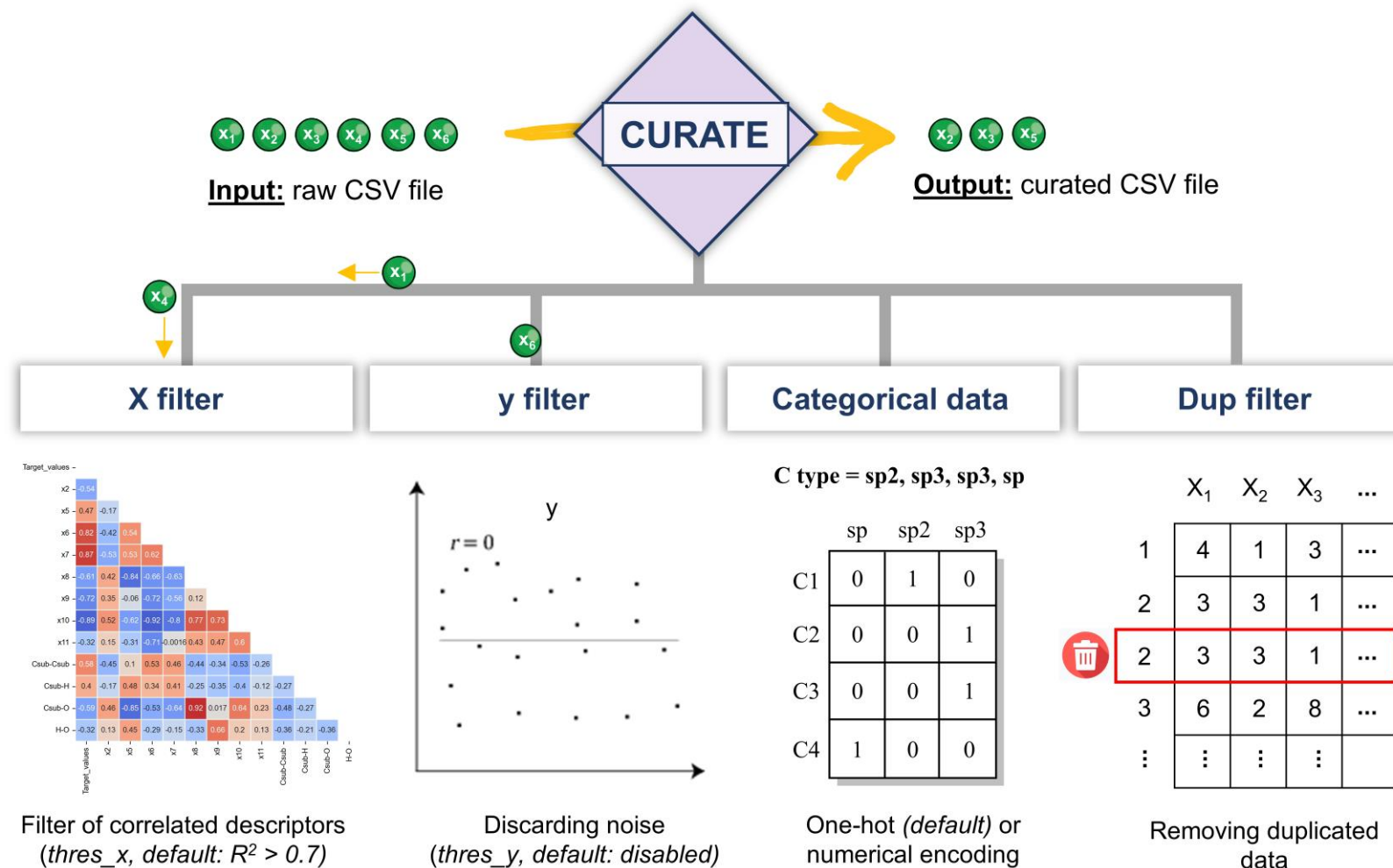
2. Filters off variables with very low correlation to the target values (**noise**)

3. Converts **categorical descriptors** into one-hot descriptors

4. Filters off **duplicates**

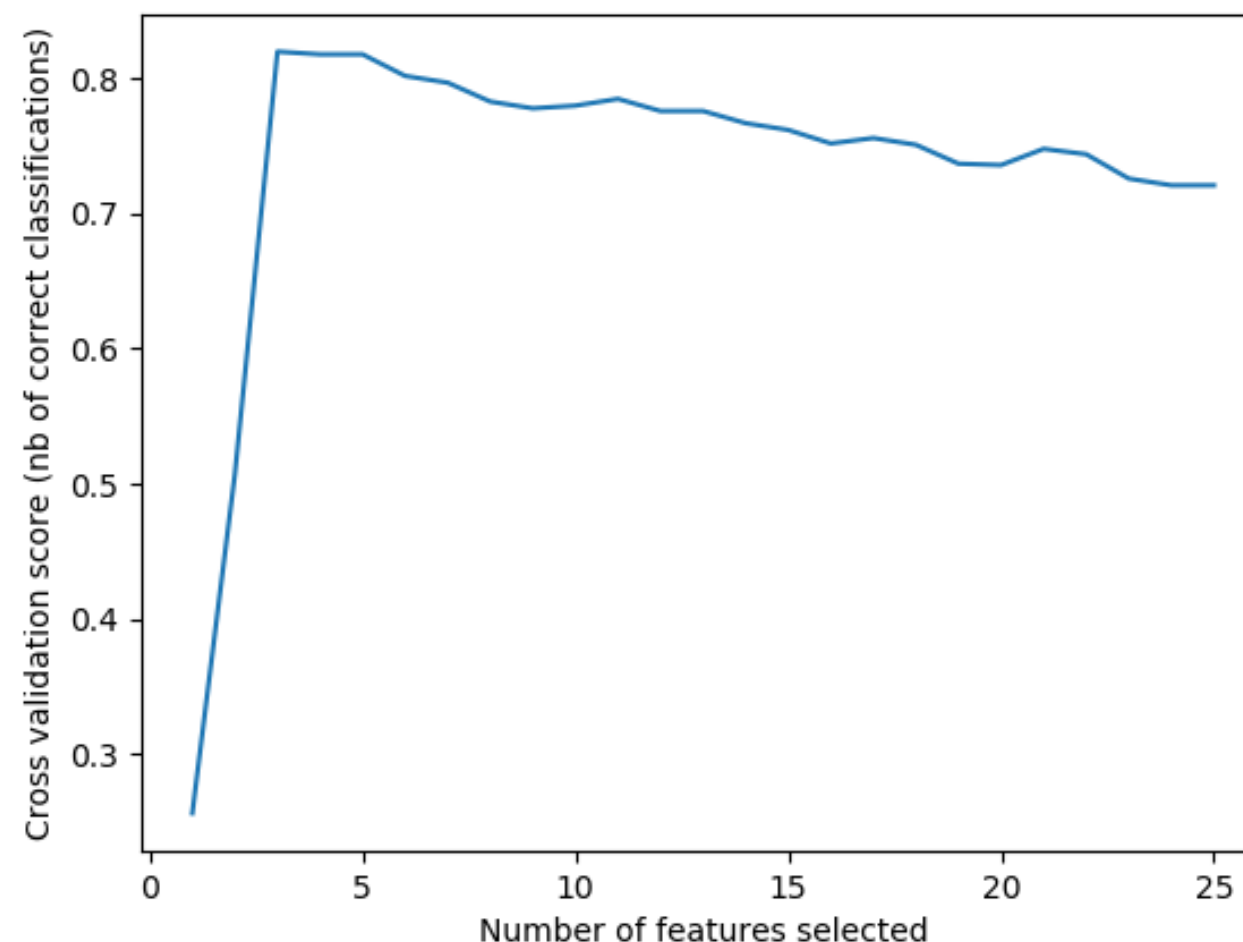
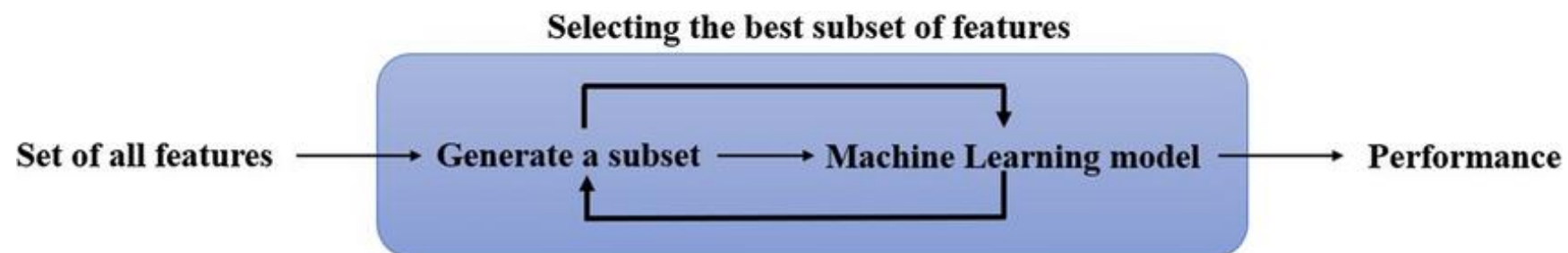
5. Filters off **missing values**
(KNN imputer, if <90% discard descriptor)

6. Keeping the **descriptors to 1/3 of the datapoints** (RFECV)



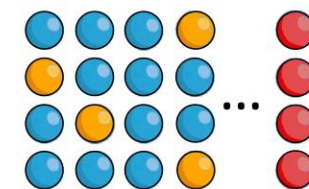
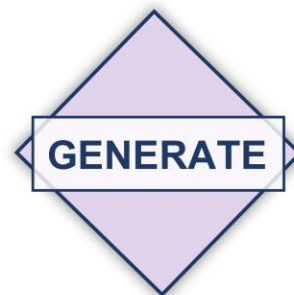
CURATE

1. Filters off **correlated descriptors**
2. Filters off variables with very low correlation to the target values (**noise**)
3. Converts **categorical descriptors** into one-hot descriptors
4. Filters off **duplicates**
5. Filters off **missing values**
(KNN imputer, if <90% discard descriptor)
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GENERATE

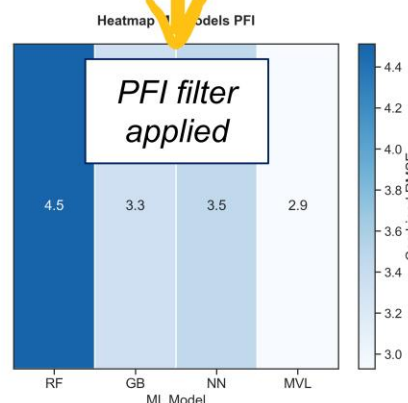
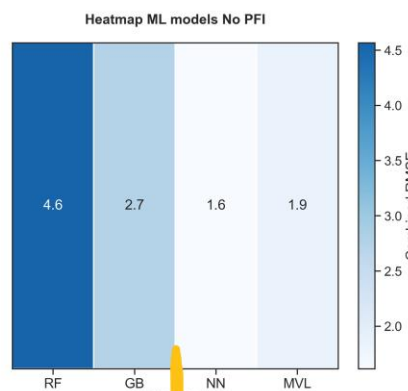
1. **Data Splitting** (Train / Validation Test)
2. **Hyperoptimizes** multiple ML models (Bayesian Optimization)
3. Filters off descriptors with low permittance feature importance (**PFI**).
4. Creates a **heatmap plot** with different model types and partition sizes.
5. Selects and stores the **best** No PFI and PFI **models**



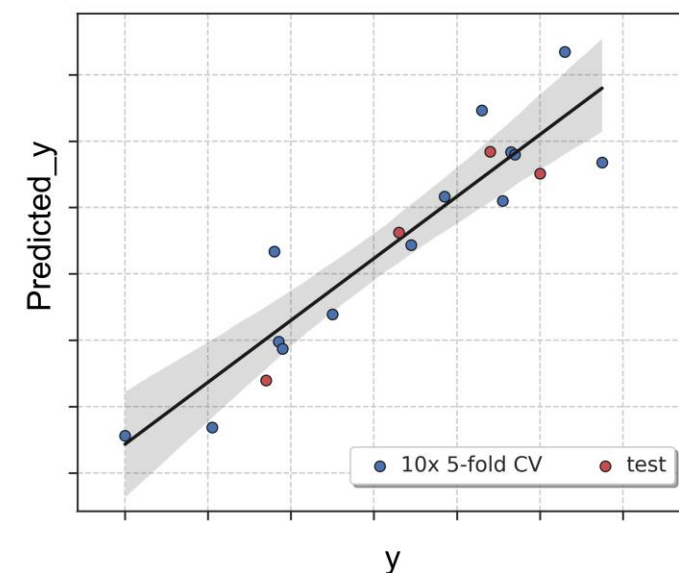
- Training set
- Validation set
- Test set (20% data)

- Data splitting: Even, random, KN, ... (default: even)
- 20% of data as test set (or 4 points at least)
- ML models included (default: model = [RF,GB,NN,MVL])
- 10 times 5-fold cross validation (Training + validation)
- Hyperoptimization of models (Bayesian optimization)
- Permutation Feature importance (default: True)

Model scans with
BO hyperoptimization

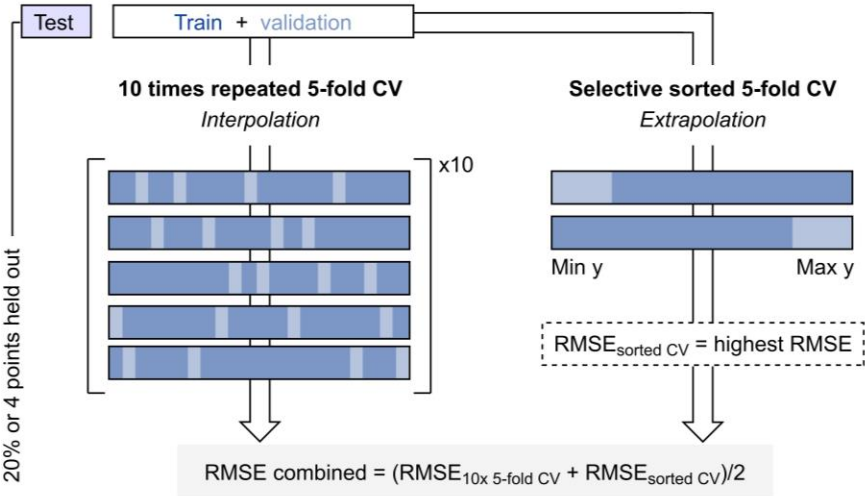


Two best models
(with and without
PFI filter)

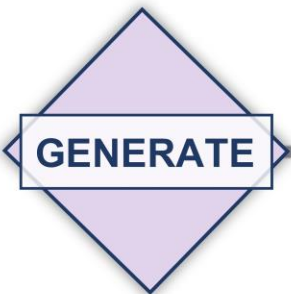
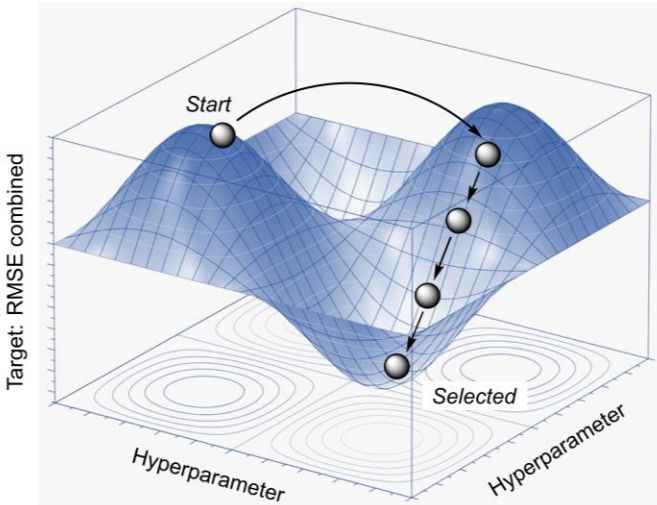


GENERATE

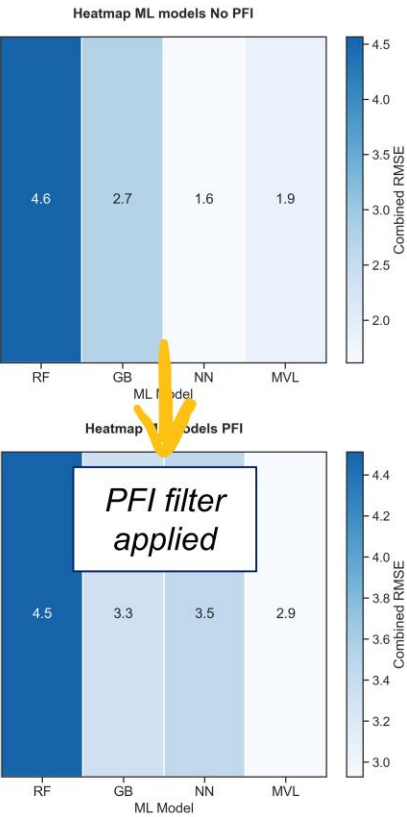
A. RMSE combined, metric sensitive to different types of overfitting



B. Bayesian hyperparameter optimization using RMSE combined



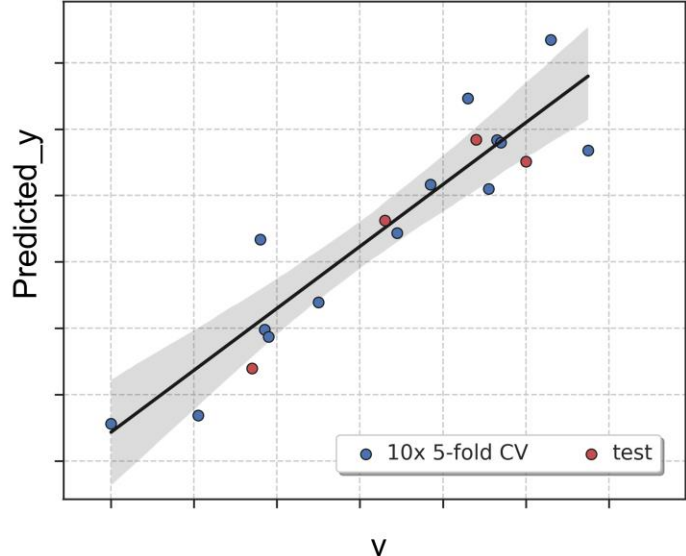
Model scans with BO hyperoptimization



Two best models (with and without PFI filter)

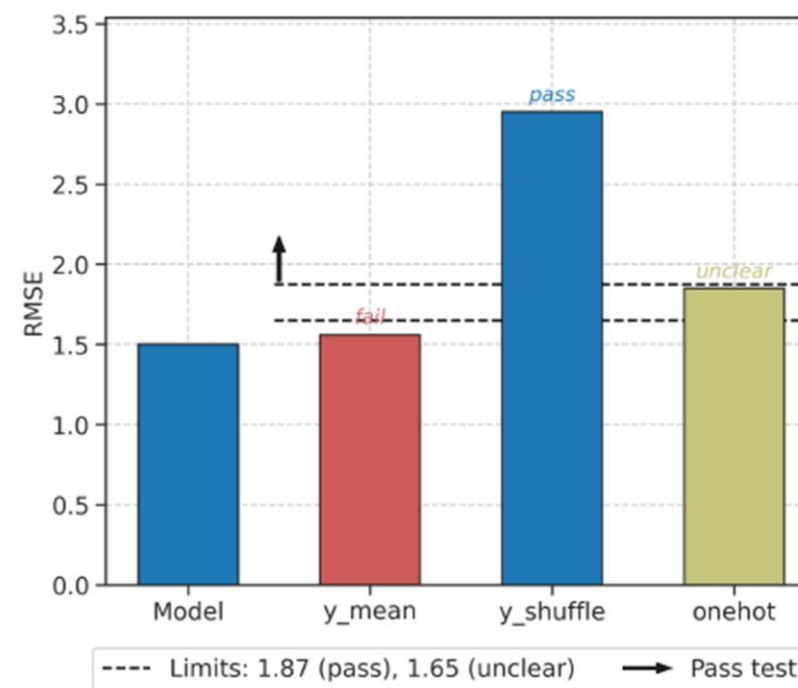
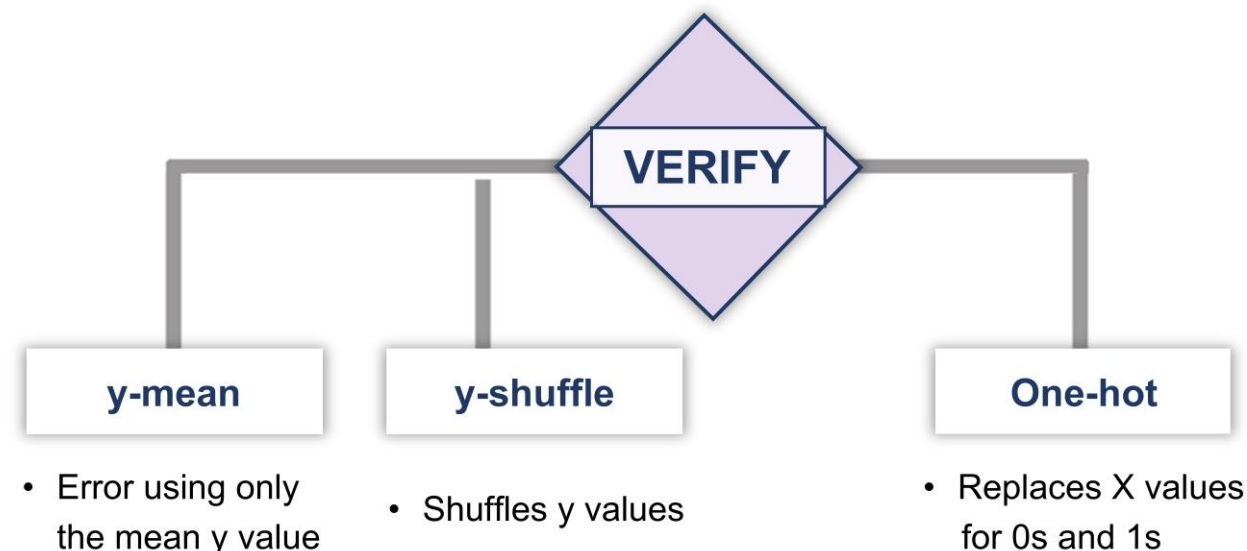
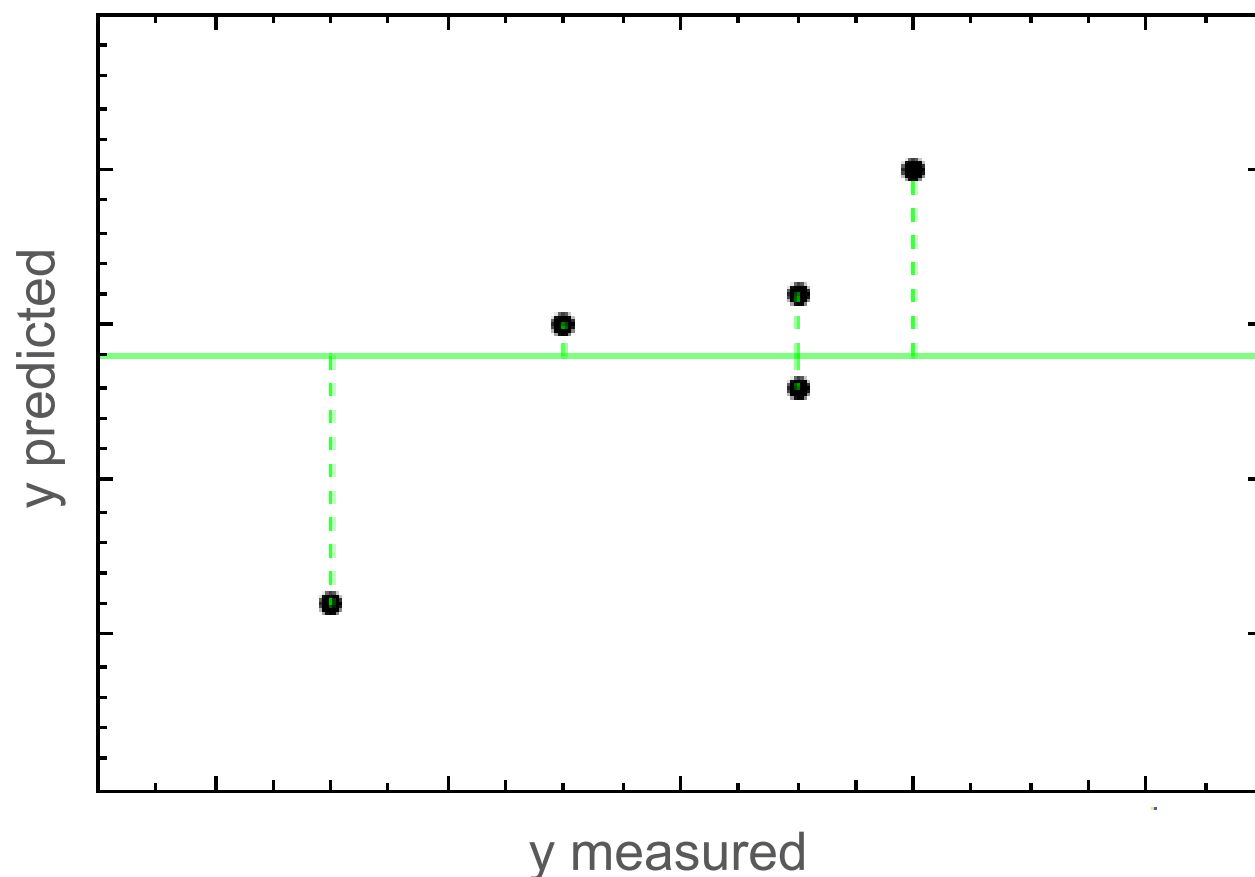
- Training set
- Validation set
- Test set (20% data)

- Data splitting: Even, random, KN, ... (default: even)
- 20% of data as test set (or 4 points at least)
- ML models included (default: model = [RF,GB,NN,MVL])
- 10 times 5-fold cross validation (Training + validation)
- Hyperoptimization of models (Bayesian optimization)
- Permutation Feature importance (default: True)



VERIFY

1. **y-mean test**: Calculates the accuracy of the model when all the predicted y values are fixed to the mean of the y values



Color codes:

- Blue: passing test
- Yellow: unclear result
- Red: failing test

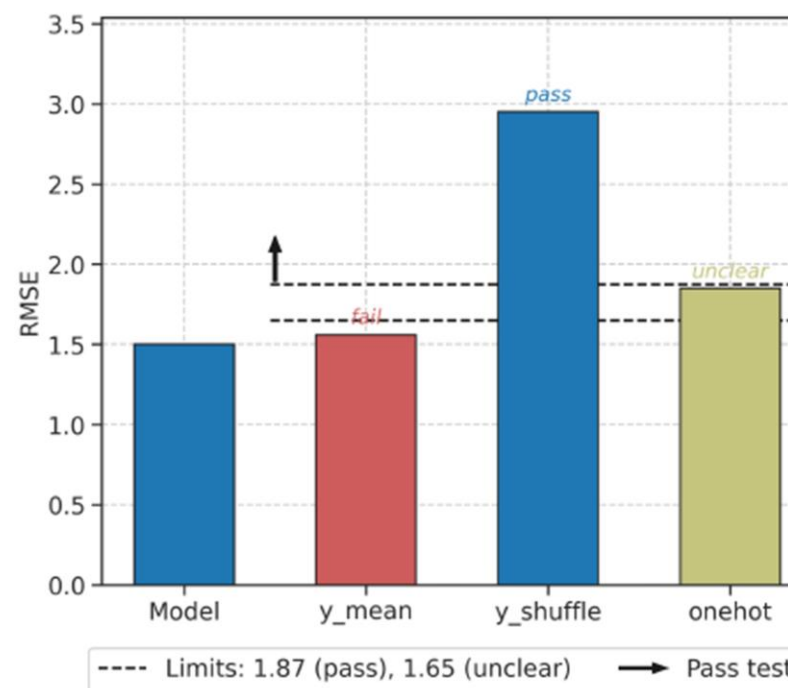
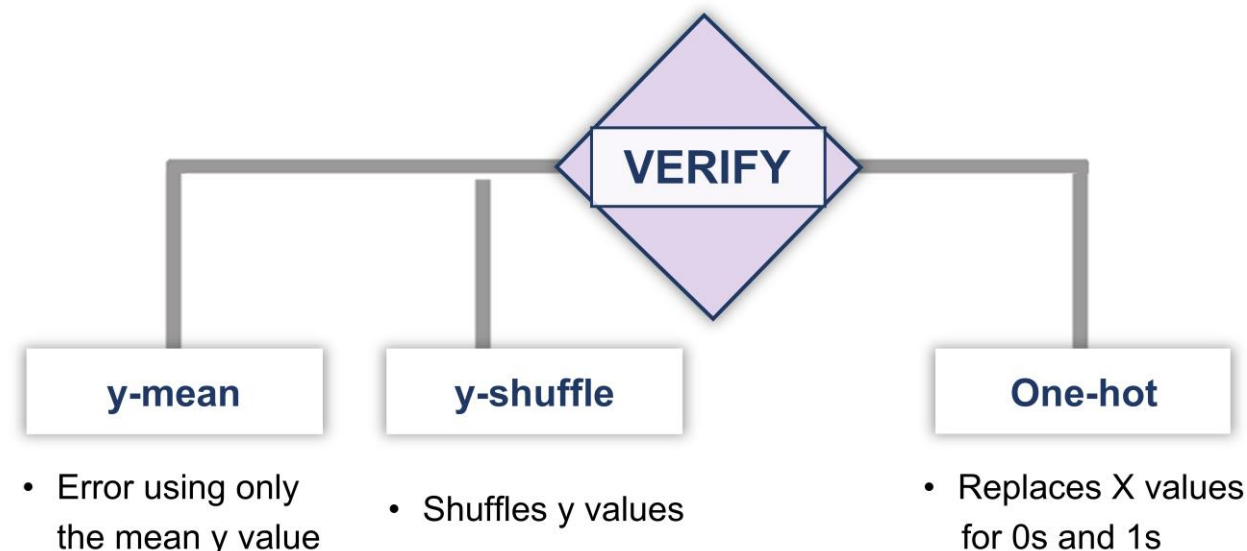
Threshold defaults:

Passing = 30 % of model
Unclear = 15 % of model

VERIFY

2. **y-shuffle test**: Calculates the accuracy of the model after shuffling randomly all the measured y values

Name	X1	y	y-shuffle
A	4,1	15	22
B	5,3	10	5
C	-2,4	5	10
D	15,1	15	2
E	-7,8	22	15
F	-0,2	2	15



Color codes:

- Blue: passing test
- Yellow: unclear result
- Red: failing test

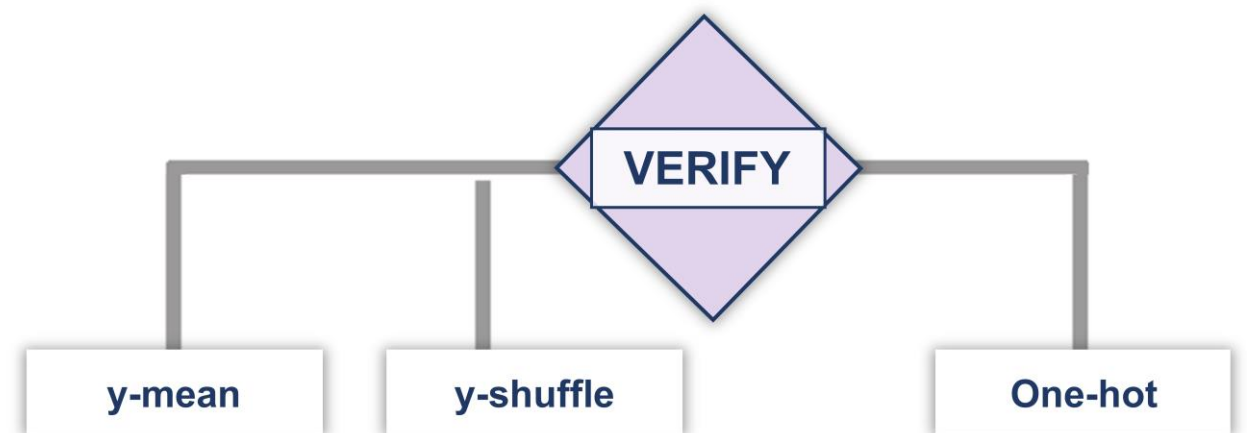
Threshold defaults:

Passing = 30 % of model
Unclear = 15 % of model

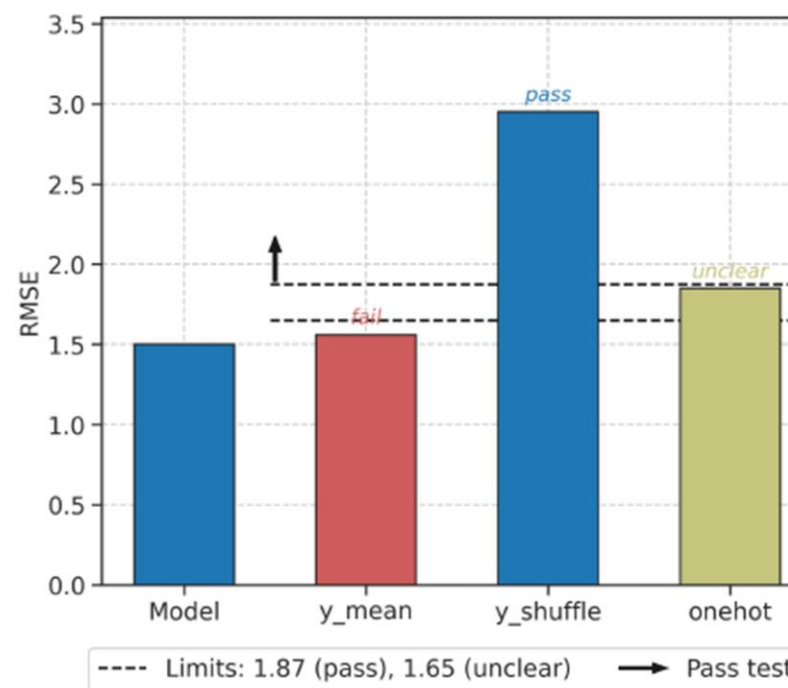
VERIFY

3. **One-hot test:** Calculates the accuracy of the model when replacing all descriptors for 0s and 1s

Name	X1	X1-OH	y
A	4,1 → 1	1	15
B	5,3 → 1	1	10
C	0,0 → 0	0	5
D	15,1 → 1	1	15
E	0,0 → 0	0	22
F	0,0 → 0	0	2



- Error using only the mean y value
- Shuffles y values
- Replaces X values for 0s and 1s



Color codes:

- Blue: passing test
- Yellow: unclear result
- Red: failing test

Threshold defaults:

Passing = 30 % of model
Unclear = 15 % of model

PREDICT

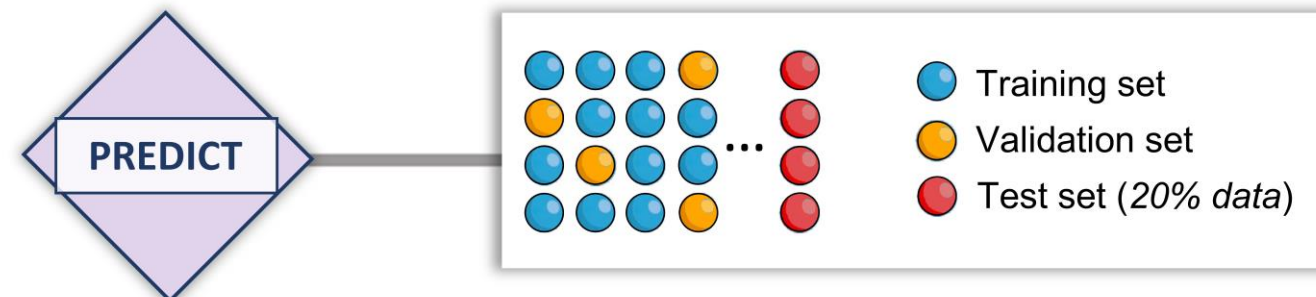
1. Calculates **R²**, **MAE** and **RMSE** (for regression) or **accuracy**, **F1** score and **MCC** (for classification)

2. Predicts values for an **external test set** (if any)

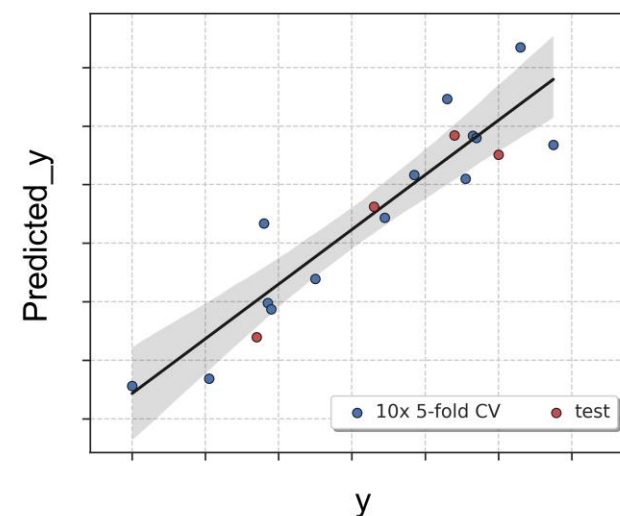
3. Performs an **outlier analysis**

4. Performs a **SHAP** feature analysis

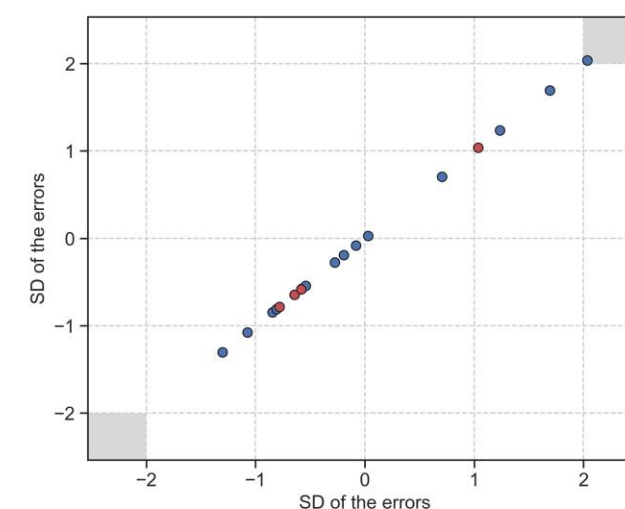
5. Performs a **PFI** feature analysis



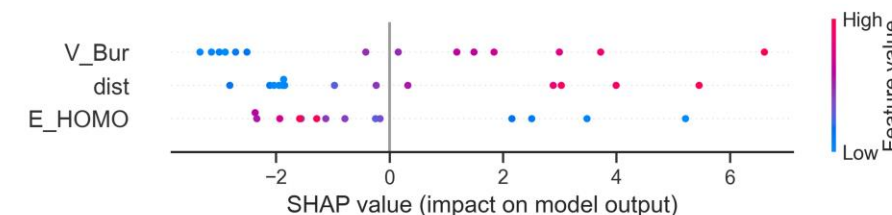
- 10x 5-fold CV (Training + validation) + test data



- Outlier analysis (default: $t_value = 2$)



- SHAP and PFI analysis





Results

- A series of folders with the names of the modules used will be created in the folder where ROBERT was executed.

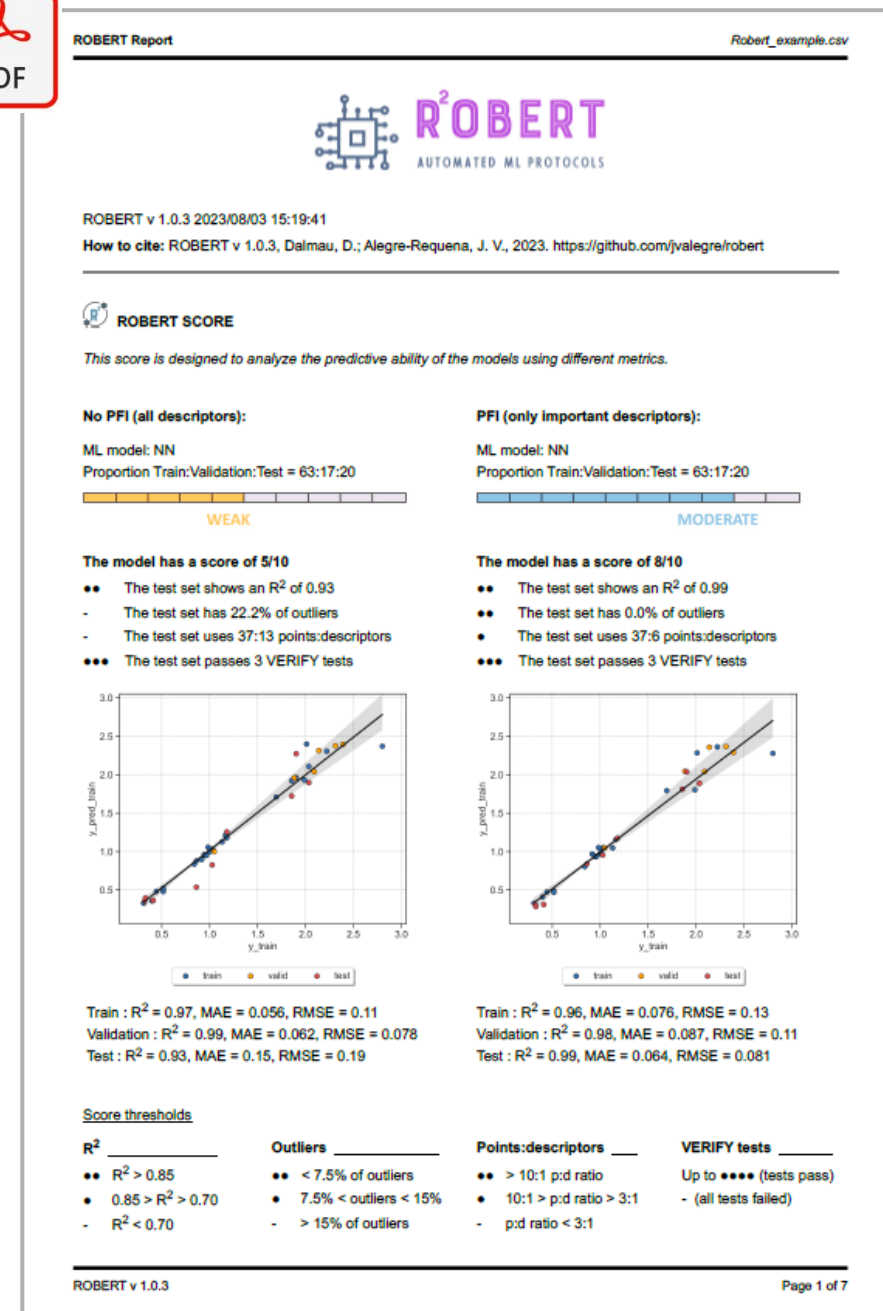
 CURATE	25/07/2023 13:47	Carpeta de archivos	
 GENERATE	25/07/2023 17:49	Carpeta de archivos	
 PREDICT	25/07/2023 11:30	Carpeta de archivos	
 VERIFY	25/07/2023 11:30	Carpeta de archivos	
 a	16/06/2023 11:32	Archivo de valores...	212 KB
 ROBERT_report	25/07/2023 11:41	Documento Adob...	1.851 KB

R²ROBERT
AUTOMATED ML PROTOCOL



- A series of folders with the names of the modules used will be created in the folder where ROBERT was executed.

- A PDF file called **ROBERT_report.pdf** will be created with the most important information



Score criteria (points)

Points	R ²	Outliers	Desc:pts
●●	> 0.85	< 7.5%	> 1:10
●	0.70 - 0.85	7.5 - 15%	1:3 - 1:10
—	< 0.70	> 15%	< 1:3

Verify tests (up to ●●●●)	● 5-fold CV ● y-mean	● y-shuffle ● One-hot
------------------------------	-------------------------	--------------------------

Points	Predictive ability
0 - 3	
4 - 6	
7 - 8	
9 -10	

No PFI (all descriptors):



Your model has a score of **5/10**

- Your model shows an R² of 0.93
- Your model has 22.2% of outliers
- Your model uses 37:13 points:descriptors
- Your model passes 3 VERIFY tests

PFI (only important descriptors):



Your model has a score of **8/10**

- Your model shows an R² of 0.99
- Your model has 0.0% of outliers
- Your model uses 37:6 points:descriptors
- Your model passes 3 VERIFY tests

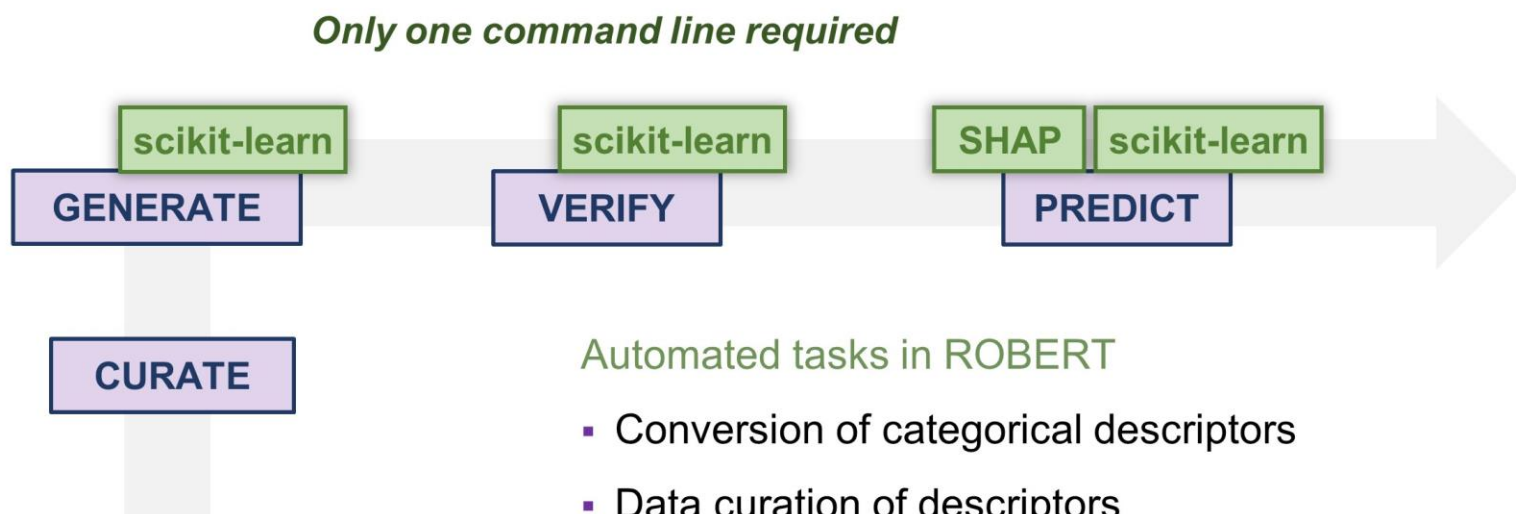
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Full workflow from csv



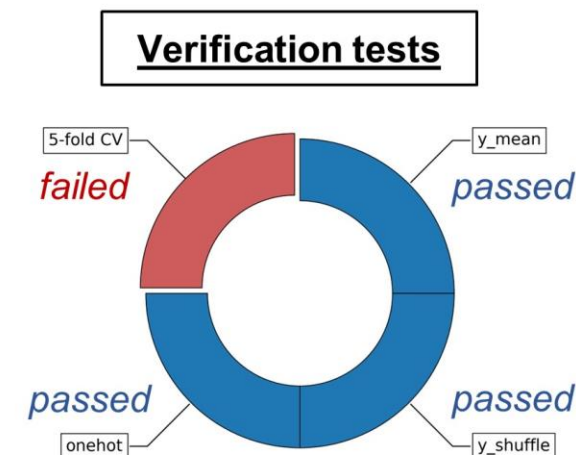
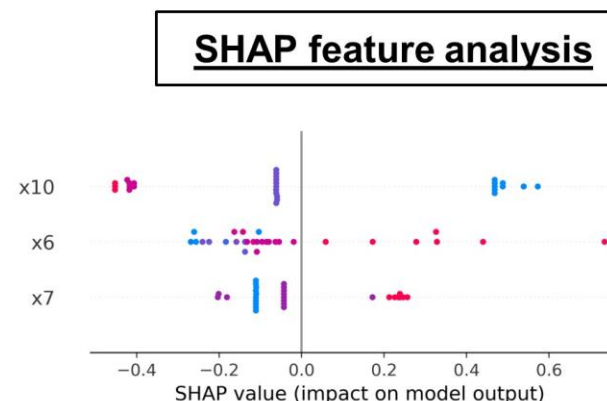
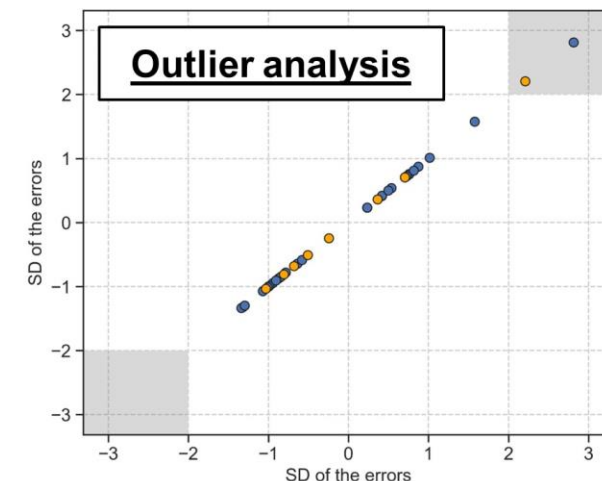
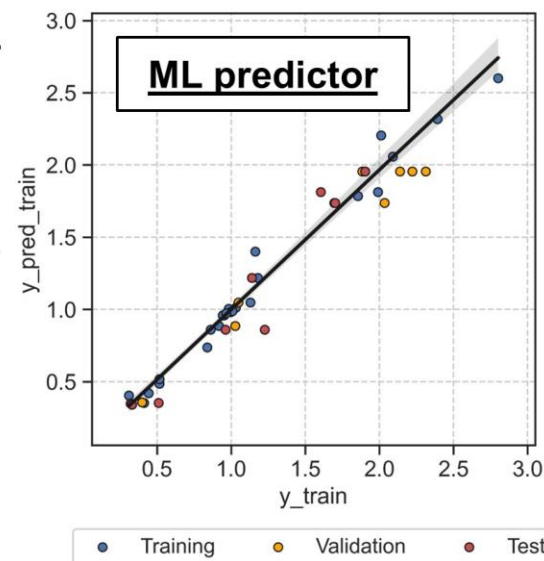
Input: CSV database

Name	y	x1	x2
Point_1	1.85	12	10.9
Point_2	2.03	11.7	10.7
Point_3	0.86	-23.05	12.1
Point_4	1.03	-16.24	12.1
Point_5	0.95	-31.94	11.9
Point_6	0.31	-68.04	13.1
Point_7	0.98	-34.71	10.8

Automated tasks in ROBERT

- Conversion of categorical descriptors
- Data curation of descriptors
- Screening of hyperoptimized ML models
- Screening of partition sizes
- New models with relevant features (PFI)
- Prediction of test set (if any) & SHAP analysis
- Analysis of outliers across different sets
- Verification tests to ensure predictive ability

Outputs:

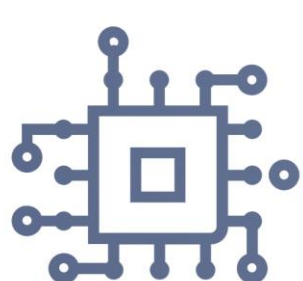


Only one command line: `python -m robert --csv_name name.csv --y target_value`

Automated generation of ML predictors



+

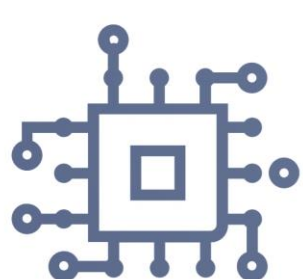


R²OBERT
AUTOMATED ML PROTOCOLS

Automated generation of ML predictors



+



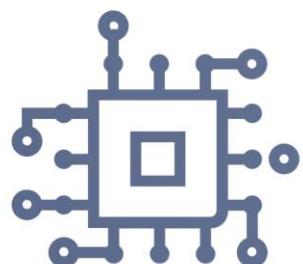
R²OBERT
AUTOMATED ML PROTOCOLS

Only one command line: `python -m robert --csv_name name.csv --y target_value --aqme`

Automated generation of ML predictors



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R²ROBERT
AUTOMATED ML PROTOCOLS

Initial AQME workflow

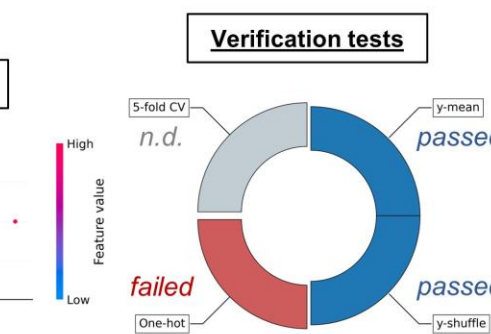
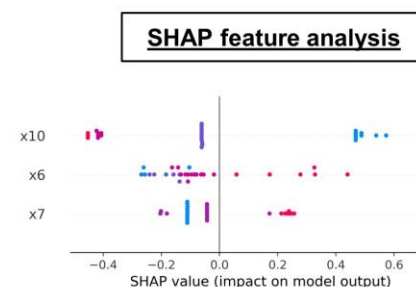
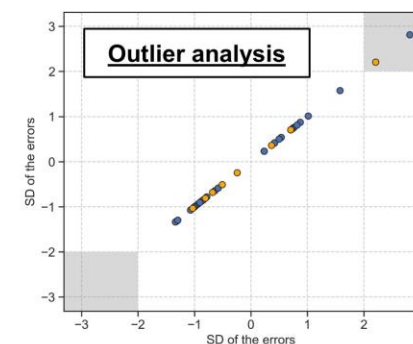
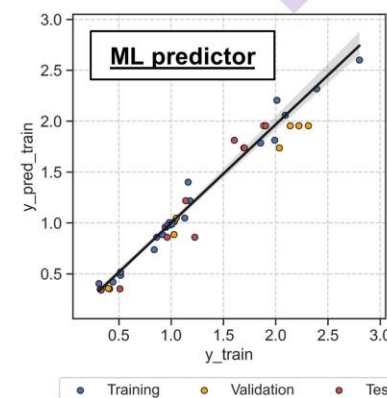
Input: CSV with SMILES

CSV database

SMILES	code_name	solub.
c1ccsc1	mol_5	-1.33
CCCC=C	mol_13	-2.68
CC(C)Cl	mol_16	-1.41
ClC(=C)Cl	mol_34	-1.64

One command line

Outputs:



Automated tasks in AQME

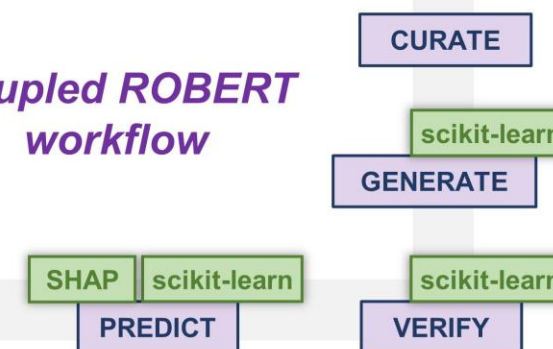
- Conformational search with RDKit
- Molecular featurization using xTB & RDKit
- Boltzmann averaging of descriptors
- Creation of a CSV database for ROBERT

- 200+ RDKit molecular descriptors
- 12 xTB molecular descriptors
- 16 xTB atomic descriptors (if any)
- DBSTEP's Buried volume (if any)

	Dipole moment	HOMO
	-0.3	-10.6
	Polarizability	LUMO
	65.1	-6.6

CSV

Coupled ROBERT workflow



Automated tasks in ROBERT

- Data curation of RDKit/xTB descriptors
- Screening of hyperoptimized ML models
- Screening of partition sizes
- New models with relevant features (PFI)
- Prediction of test set (if any) & SHAP analysis
- Analysis of outliers across different sets
- Verification tests to ensure predictive ability

Only one command line: `python -m robert --csv_name name.csv --y target_value --aqme`



Automated generation of ML predictors

ESOL: Estimating Aqueous Solubility Directly from Molecular Structure

John S. Delaney

View Author Information ▾

✓ **Cite this:** *J. Chem. Inf. Comput. Sci.* 2004, 44, 3, 1000–1005

Publication Date: March 13, 2004 ▾

<https://doi.org/10.1021/ci034243x>

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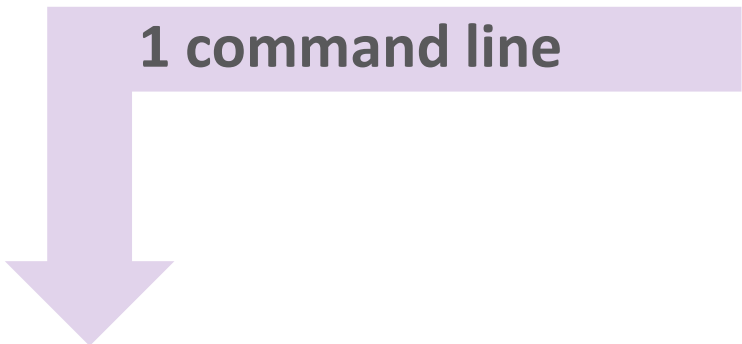
Supporting Info (1) »

SUBJECTS: [Molecular modeling](#), [Molecules](#), ▾



Automated generation of ML predictors

1 command line



ESOL: Estimating Aqueous Solubility Directly from Molecular Structure

John S. Delaney

View Author Information ▾

✓ **Cite this:** *J. Chem. Inf. Comput. Sci.* 2004, 44, 3, 1000–1005

Publication Date: March 13, 2004 ▾

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Supporting Info (1) »

SUBJECTS: Molecular modeling, Molecules, ▾

Automated generation of ML predictors

1 command line

ESOL: Estimating Aqueous Solubility Directly from Molecular Structure

John S. Delaney

View Author Information

Cite this: *J. Chem. Inf. Comput. Sci.* 2004, 44, 3, 1000–1005

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Supporting Info (1) »

SUBJECTS: Molecular modeling, Molecules, »

A Estimating aqueous solubility

Input: CSV with SMILES

ESOL database

SMILES	code_name	solub.
<chem>c1ccsc1</chem>	mol_5	-1.33
<chem>CCCC=C</chem>	mol_13	-2.68
<chem>CC(C)Cl</chem>	mol_16	-1.41
<chem>ClC(=C)Cl</chem>	mol_34	-1.64

AQME

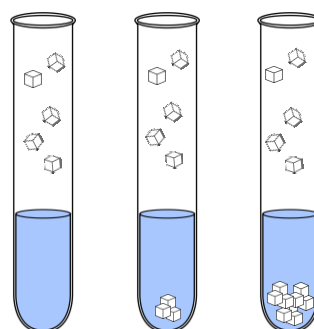
CCCC=C mol_13

Conformer sampling

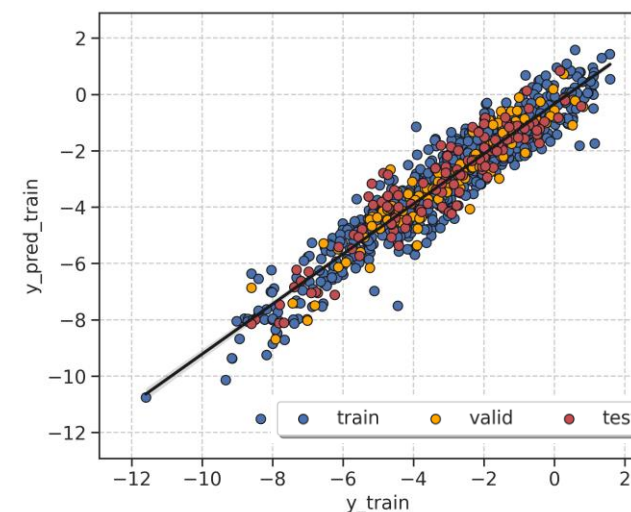
Mol. featurization

200+ RDKit, xTB and DBSTEP descriptors

ROBERT



Solubility prediction



ROBERT SCORE

ML model: MVL

Proportion Train:Validation:Test = 81:9:10



The model has a score of 10/10

- The test set shows an R^2 of 0.89
- The valid. set has 4.9% of outliers
- Using 1012:47 points(train+valid.):descriptors
- The valid. set passes 4 VERIFY tests

Only one command line: `python -m robert --csv_name solubility.csv --y solubility --aqme --qdescp_keywords "--qdescp_solvent h2o"`

Automatic generation of predictors

Input: CSV with SMILES

SMILES	code_name	$\Delta E^\ddagger$
[Ir]([P+](CC)...	ir_tbp_1_dft-pet3...	8.9
[Ir]([P+](C)...	ir_tbp_1_dft-pme3...	6.5
[Ir]([As+](C)...	ir_tbp_1_dft-asme3...	17.9
[Ir]([n+]1ccn...	ir_tbp_1_dft-pyz...	22

Machine learning dihydrogen activation in the chemical space surrounding Vaska's complex†‡



[Pascal Friederich](#),  ^{abc} [Gabriel dos Passos Gomes](#),  ^{ac} [Riccardo De Bin](#), ^d [Alán Aspuru-Guzik](#) ^{*acef}
and [David Balcells](#)  ^{*g}

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[Ir]([n+]1ccn...	ir_tbp_1_dft-pyz...	22

1 command line



**Machine learning dihydrogen
activation in the chemical space
surrounding Vaska's complex†‡**



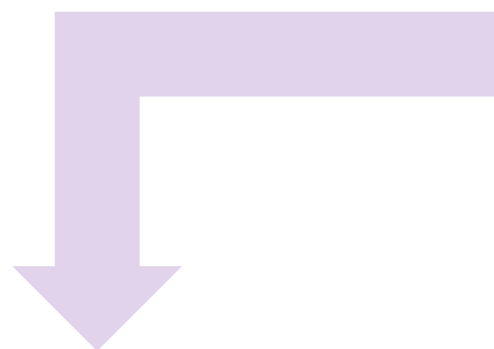
[Pascal Friederich](#),  ^{abc} [Gabriel dos Passos Gomes](#),  ^{ac} [Riccardo De Bin](#), ^d [Alán Aspuru-Guzik](#) ^{*acef}
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Machine learning dihydrogen activation in the chemical space surrounding Vaska's complex ‡



Pascal Friederich, ^{abc} Gabriel dos Passos Gomes, ^{ac} Riccardo De Bin, ^d Alán Aspuru-Guzik ^{*acef}
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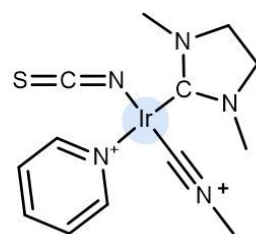
B Application to catalysis: Estimating activation energies

Input: CSV with SMILES

Vaska's complex database

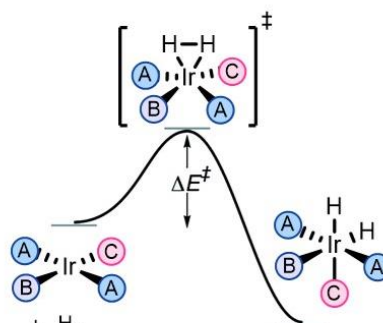
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[Ir]([n+]1ccn...	ir_tbp_1_dft-pyz...	22

Step 1: AQME

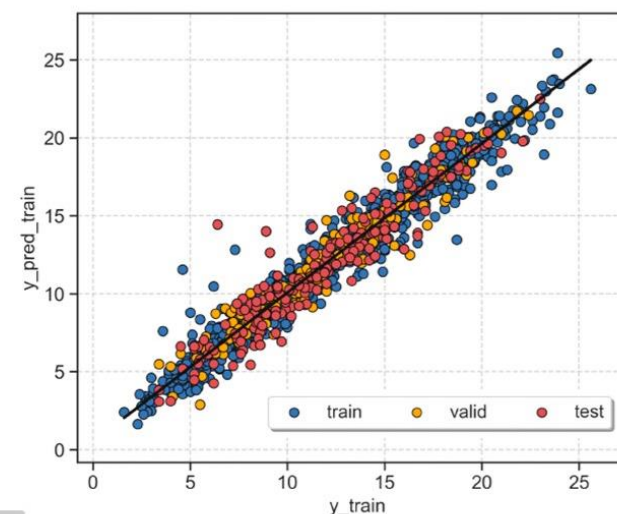


200+ *electronic & steric*
atomic/molecular descps.
(RDKit, xTB & DBSTEP)

Step 2: ROBERT



$\Delta E^\ddagger$ prediction



ROBERT SCORE

ML model: NN
Proportion Train:Validation:Test = 81:9:10



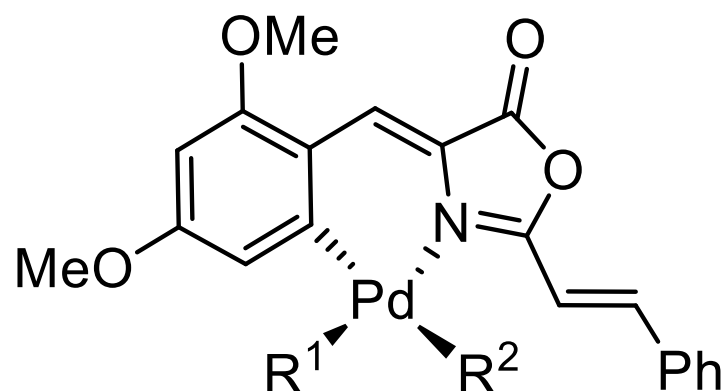
The model has a score of 9/10

- The test set shows an R^2 of 0.89
- The valid. set has 10.5% of outliers
- Using 1711:21 points(train+valid.):descriptors
- The valid. set passes 4 VERIFY tests

Only one command line: `python -m robert --csv_name vaskas.csv --y barrier --aqme --qdescp_keywords "--qdescp_atoms [Ir]"`

Real example of our lab

- Discovery of new Pd-oxazolone luminiscent compounds



1a: R¹ = Pyr, R² = TFA (18% QY)

1b: R¹ = Pyr, R² = Cl (12%)

1c: R¹ = Pyr, R² = Pyr (28%)

1d: R¹ = NHC, R² = TFA (15%)

1e: 8-Aminoquinoline (12%)

Eur. J. Org. Chem., **2018**, 6158-6166.

Molecules **2021**, 26, 1238.

Org. Biomol. Chem., **2021**, 19, 2773-2783

Inorg. Chem. **2023**, 62, 9792–9806.

From all the compounds reported across four articles, only five examples with QY > 10%!

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From all the compounds reported across four articles, only five examples with QY > 10%!



We could go to the lab and try new possible combinations



Experim.

Real example of our lab

- Discovery of new Pd-oxazolone luminiscent compounds

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Molecules **2021**, 26, 1238.

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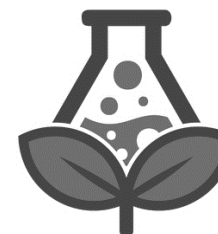
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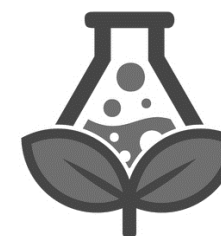
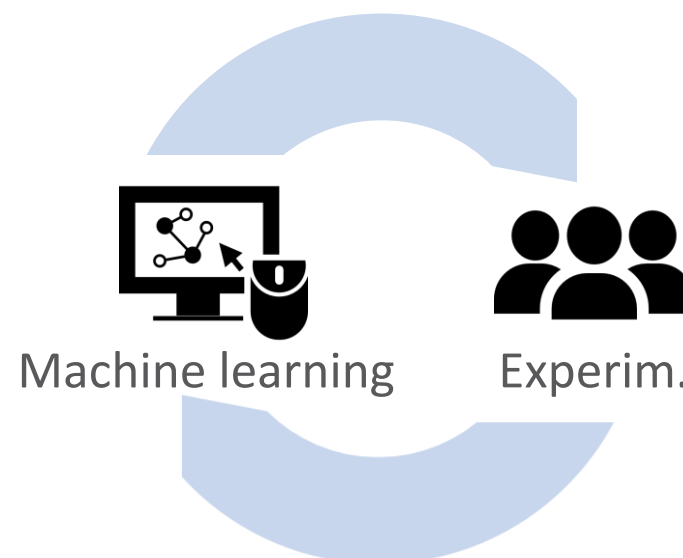
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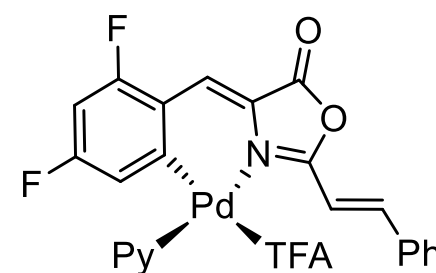
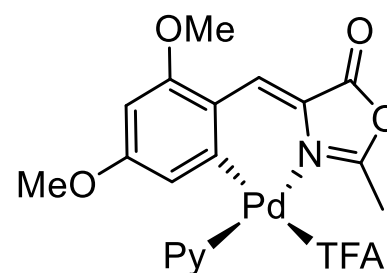
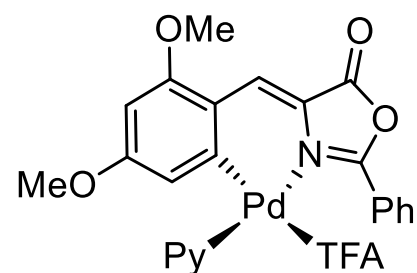
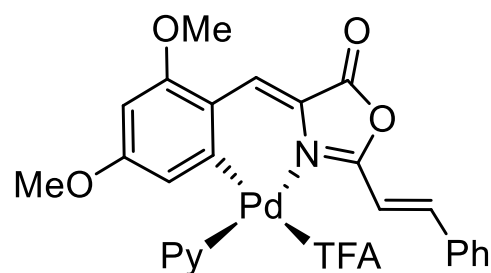


~~We could go to the lab and try each of the combinations~~

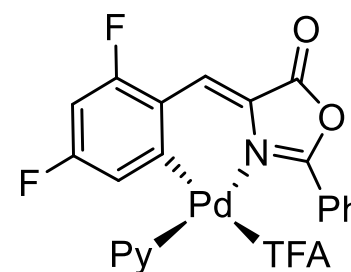
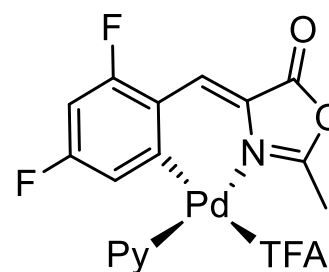
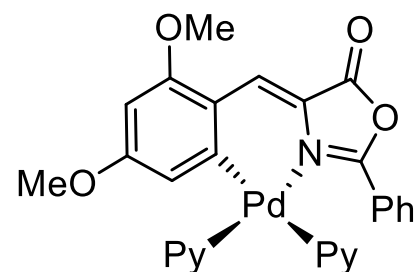
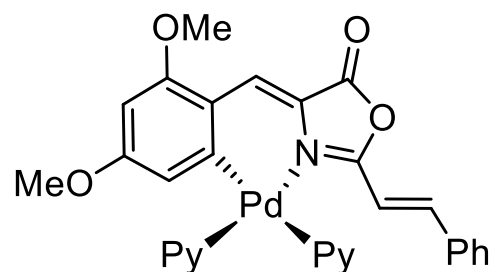


Real example of our lab

1. Draw in ChemDraw the 23 compounds that already exist

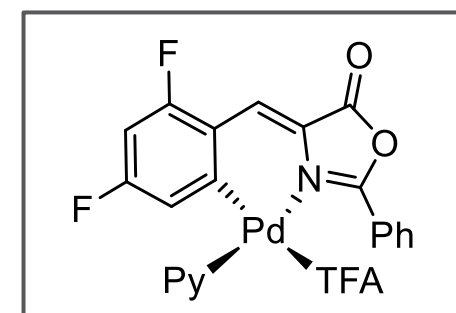
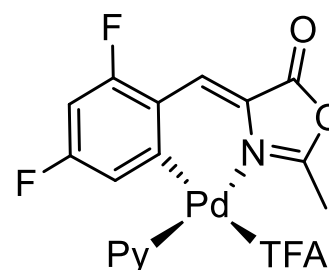
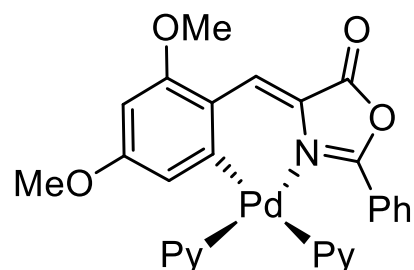
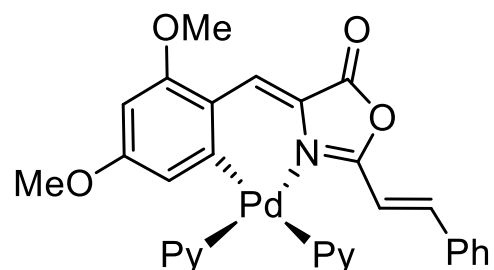
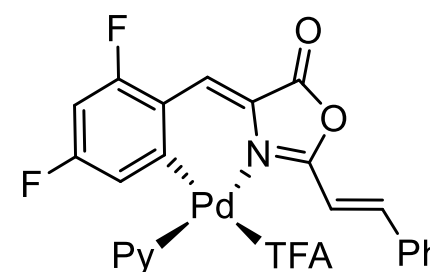
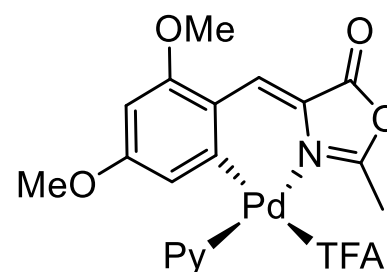
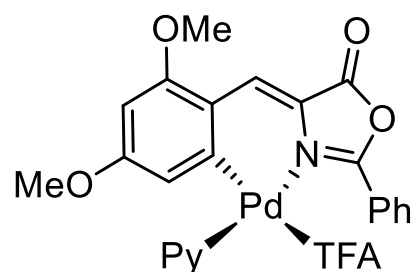
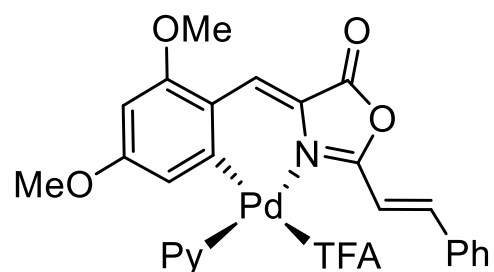


...



Real example of our lab

1. Draw in ChemDraw the 23 compounds that already exist



ccc(F)cc(F)C=C...

2. Convert the structures to SMILES strings (Ctrl + Alt + C)

Real example of our lab

3. Create the Excel database with names, SMILES and QY

code_name	SMILES	QY(%)
complex_1	<chem>O=C1OC=N/C1=C\C2...</chem>	2
complex_2	<chem>[Pd]C1=CC(OC)=CC(OC)...</chem>	5
complex_3	<chem>FC1=C2C([Pd][N]C=C2)...</chem>	12
...		

Real example of our lab

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...		

4. Run ROBERT

```
> python -m robert --y QY(%) --names code_name --aqme --csv Pd.csv
```

Real example of our lab

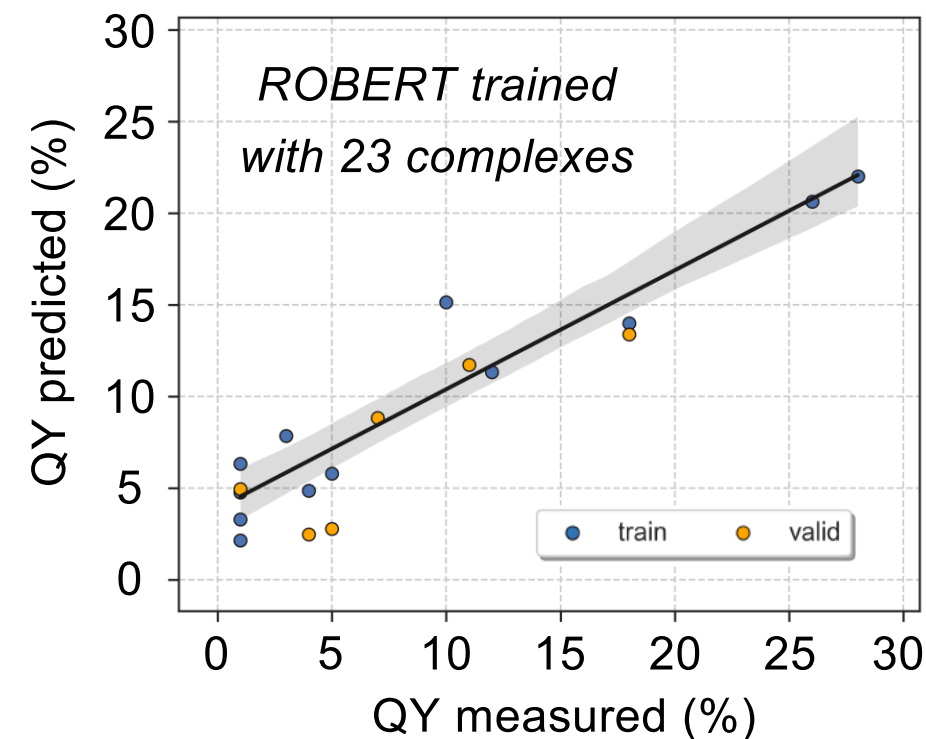
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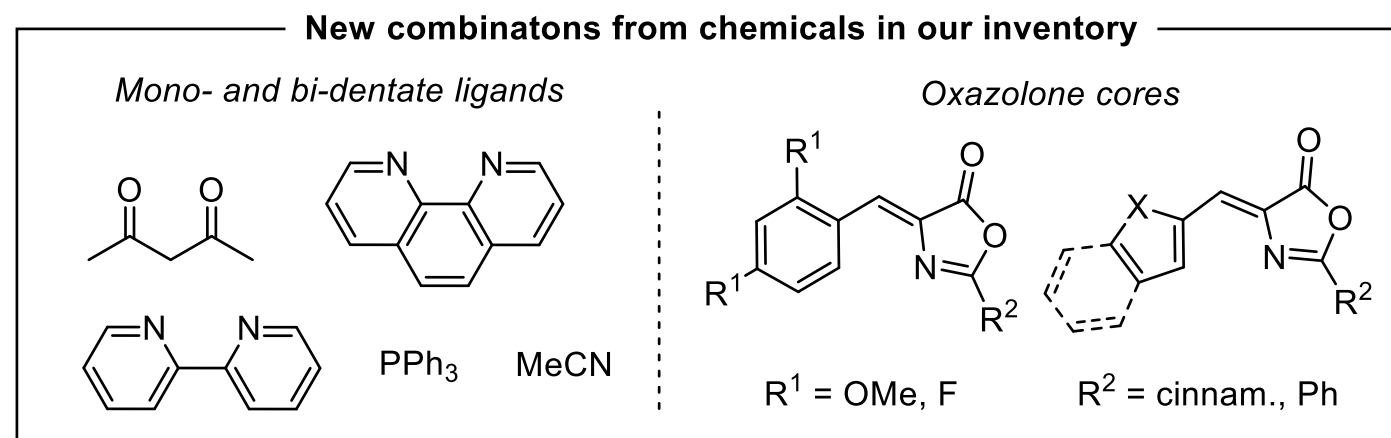
```
> python -m robert --y QY(%) --names code_name --aqme --csv Pd.csv
```

Aprox. 30-45 min



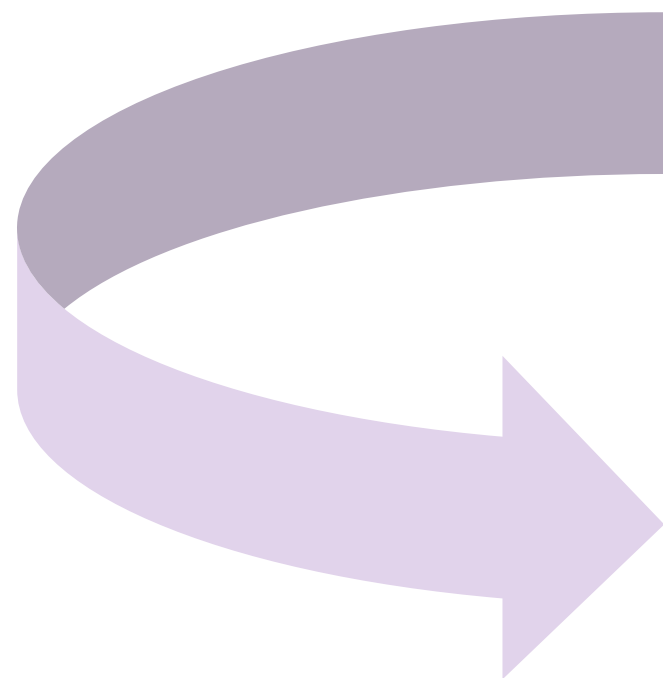
Real example of our lab

- ROBERT for predicting new candidates



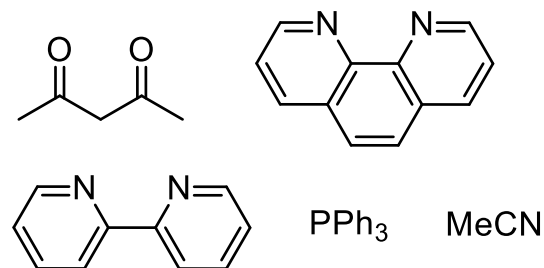
Real example of our lab

- ROBERT for predicting new candidates

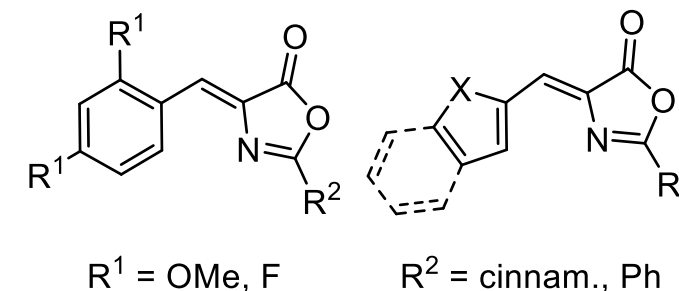


New combinations from chemicals in our inventory

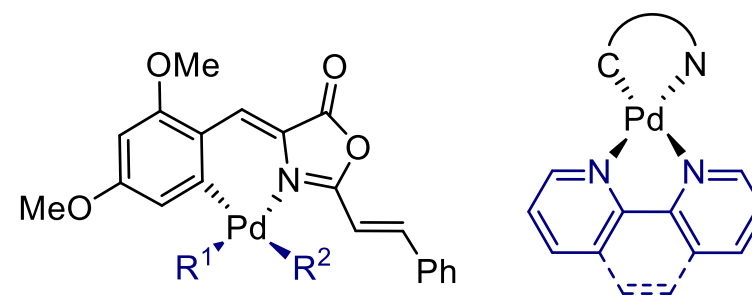
Mono- and bi-dentate ligands



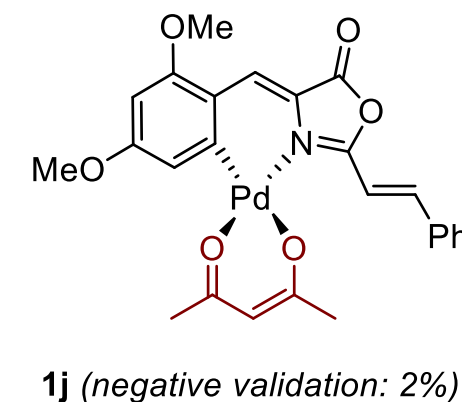
Oxazolone cores



Four ROBERT candidates



11 ROBERT discards



Real example of our lab

- ROBERT for predicting new candidates



- Lab-tested: 80% Accuracy

